Alex Castillo González

Applied AI Engineer · LLM integration and agent systems

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SUMMARY

Applied AI engineer on the **LLM integration layer** — streaming, tool calls, usage accounting, provider changes and model retirements — and on the agent systems that do that work at scale. I designed and run a multi-agent pipeline that finds, reproduces and fixes bugs in open-source AI tooling, with an independent reviewer gating every change. Maintainers have merged **736 of its pull requests across 169 projects** since August 2026 (87% of those decided; median 18 hours), 167 in projects with over 10k★, including **semantic-kernel**, **deer-flow**, **LiteLLM**, **MLflow**, **redis-py**, **Sentry** and **Haystack**. I decide where to work, set the quality bar and keep cost per change down; the fixes became `tanda` and `llm-drift-lint`. Before that I founded and ran a business on my own in Toronto.

SKILLS

LLM integration — Anthropic, OpenAI (Chat Completions and Responses), Gemini / Vertex, Bedrock, Mistral and OpenAI-compatible APIs · **streaming and parallel tool calls** · **token, cache and reasoning-token accounting**, cost tracking · finish and stop reasons, refusals, thinking blocks · model migrations and retirements · **Model Context Protocol (MCP)** servers and clients · tracing and GenAI instrumentation (OpenTelemetry, Logfire, Phoenix, Opik, MLflow)

Agent systems — multi-agent pipelines with coding agents (Claude Code, Codex, opencode) · role design, shared work queues, **independent review gates** · cost and context budgeting per role and per model · triage, volume limits and per-repo policies · evaluation of agent output against tests and real merges

Languages & backend — Python · pytest · asyncio · Ruby on Rails · Go, C# and TypeScript codebases (reading, fixing, testing upstream) · PostgreSQL · REST APIs

Web & infra — TypeScript · Next.js · React · Tailwind · **AWS** (Lambda, DynamoDB, ECR, IAM) · **Terraform** · CI/CD with GitHub Actions and OIDC · Docker · Vercel · Git

EXPERIENCE

Open-source engineering with an agent pipeline — Designer & operator · Remote
Aug 2026 – present

- Designed a pipeline of coding agents around one shared queue: scouts find bugs users hit in current `main`, fixers reproduce and fix them with a failing-first test, **independent reviewers on a stronger model** re-run the test before anything is published, and a follow-up role answers maintainer reviews.
- Set its rules: only work maintainers want, per-repo caps, skip repos that reject AI-assisted contributions, disclose AI involvement. Tracked cost, merge rate and context per run: 88% 30-day merge rate, about $2.50–4 of model usage per published PR.
- 736 merged pull requests across 169 projects, 167 in projects with over 10k★.

Churrería Calderón — Founder & owner-operator · Toronto, Canada
Sep 2025 – Jul 2026 *(business closed July 2026)*

- Moved my business from Spain to Toronto on my own: leased premises in Kensington Market, imported the machinery, registered the business and obtained every permit to operate legally. Opened in December 2025 and ran it with no hires — production, service, purchasing, cash and compliance — until it closed in July 2026.

Family churrería business — Manager · Catalonia, Spain
2023 – Aug 2025 · mainly summer seasons *(and in the same business before that)*

- Managed the family business, mainly in the summer seasons — production, service, purchasing and cash — and took it from a mobile trailer to fixed premises.

Le Wagon — Part-time Programming Teacher · Remote (France)
Oct 2022

- Taught the part-time flex cohort of the fullstack bootcamp — Ruby, OOP, SQL and PostgreSQL, HTML/CSS/JavaScript, Rails — through exercises and live debugging.

TECNOBIT (Grupo Oesía) — Fullstack Developer · Valdepeñas, Spain
Aug 2022 – Nov 2022 · On-site

- Shipped features into a long-running internal application maintained by a team of five: Ruby routines pulling data from JSON files and the **Jira API** into PostgreSQL with multi-stage validation, and role-dependent views that surfaced subprocess state.

PROJECTS

tanda — the pipeline, as a public tool

Code

Runs a fleet of coding agents (Claude Code, opencode, Codex or any CLI) around one SQLite queue, with a review gate and a local dashboard. Each role may only set the verdicts its config allows; an agent that exits without a verdict sends its item to a human. Waiting is done by tanda, not by the agents, so idle roles cost no tokens.

llm-drift-lint — static checks for LLM calls that break when providers change

Code

Finds Anthropic, OpenAI and Gemini SDK calls in Python code, resolves the model id and checks each call against what that model accepts today: sampling parameters newer models reject, `content[0].text` when the first block is now a thinking block, a forced `tool_choice` under thinking, defaults past their shutdown date.

OPEN SOURCE

736 merged pull requests across 169 open-source projects since August 2026, 87% of those decided; median 18 hours to merge; 40 projects over 10k★. The same bugs recur across the ecosystem — streamed parallel tool calls merged when a provider omits the index (fixed in 24 repositories), cache and reasoning tokens missing from usage, sampling parameters sent to models that reject them. Examples:

- **LiteLLM** (#44795): streamed Vertex AI pass-through calls were priced without their location; both modes now price the same way.
- **LLM Tornado** (#205): Anthropic streaming re-yielded every tool call at each block end, so three parallel tools ran six times. 88 merged there.
- **Haystack** (#12662): `run_async` had drifted from `run` — no concurrency limit, a lock shared across event loops. 14 merged (write-up).
- **deer-flow** (#7677, 85k★) and **semantic-kernel** (#14700): a tool argument swallowed by runtime injection; embeddings batched to the API limit.

Full searchable log: portfolio-alexgonzalez33.vercel.app.

EDUCATION

Le Wagon — Fullstack Web Development bootcamp, 2022. Self-directed study since.